OPSINTEL AI

Evidence-based pipeline investigation

From failed overnight run to traceable next action - powered by Exasol Personal

EXATHON INDIA 2026 / TRACK 04: THE WORK BEHIND THE WORK

Public prototype • MIT licensed • Synthetic data

THE PROBLEM

The failure is visible. The explanation is scattered.

MANUAL

Evidence lives in separate views

The engineer finds the run, compares row counts, searches logs, and decides what to inspect next.

OPSINTEL

01 Select the failed run

02 Review cited status, counts, and logs

03 Compare candidate explanations

04 Human verifies before action

Expected benefit: fewer context switches. The impact study will measure time and evidence selection.

PRODUCT WORKFLOW

Operational evidence flows through Exasol, FastAPI, and the dashboard

01 **Pipeline runs and logs** →

02 **Exasol SQL analytics** →

03 **FastAPI rules and candidates** →

04 **Dashboard + optional LLM narrative** →

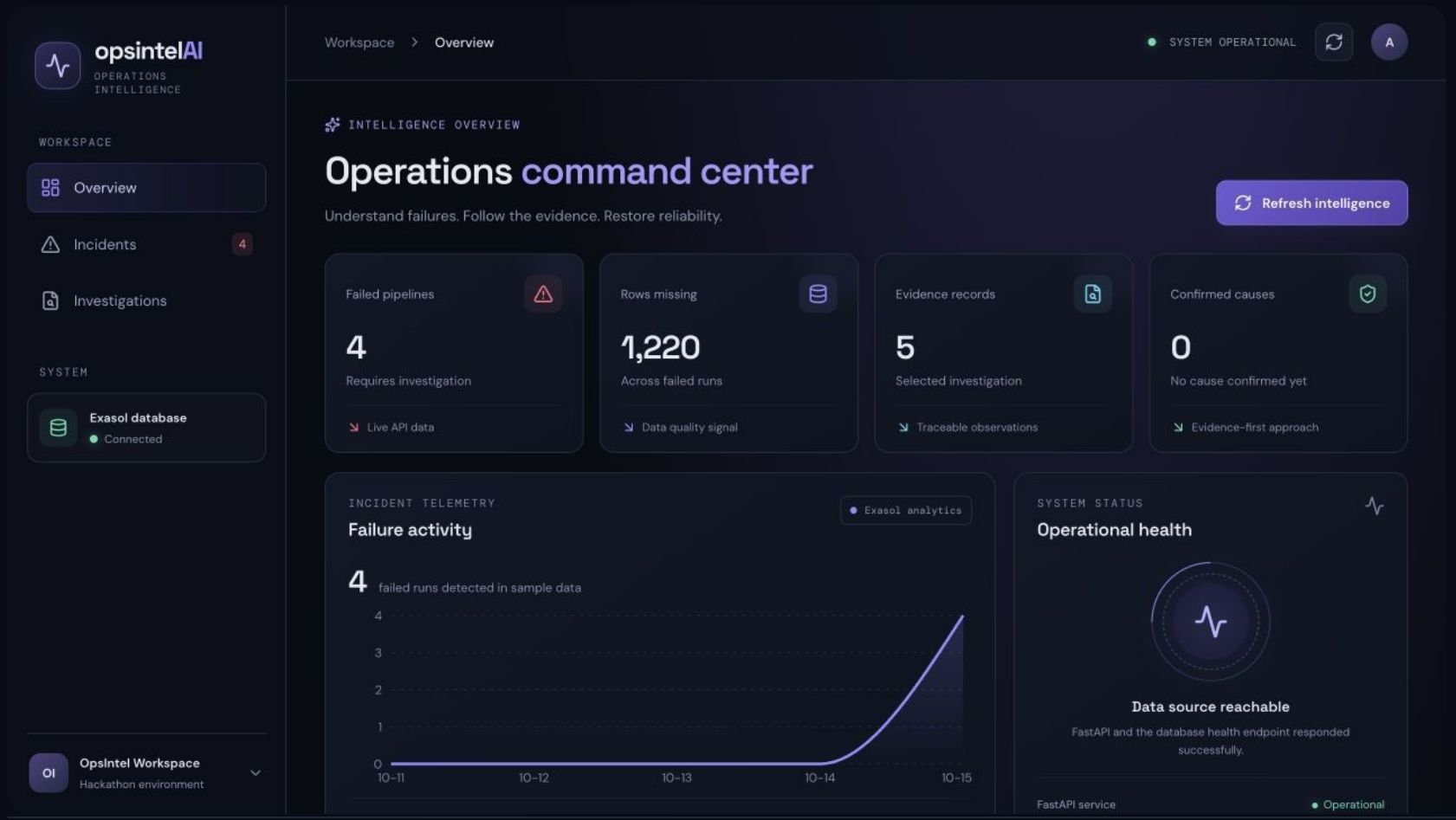
05 **Engineer verifies and confirms**

Safety boundary

Rules produce the category and candidates. The optional LLM only summarizes retrieved evidence. Confirmation and repairs stay human-controlled.

LIVE PRODUCT

The dashboard reads incident telemetry calculated in Exasol



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failed pipeline runs

1,220

aggregate missing rows across four demo incidents

✓ Exasol connected
✓ Trend from SQL view
✓ Incidents selectable

INVESTIGATION EXPERIENCE

Every recommendation points back to source evidence

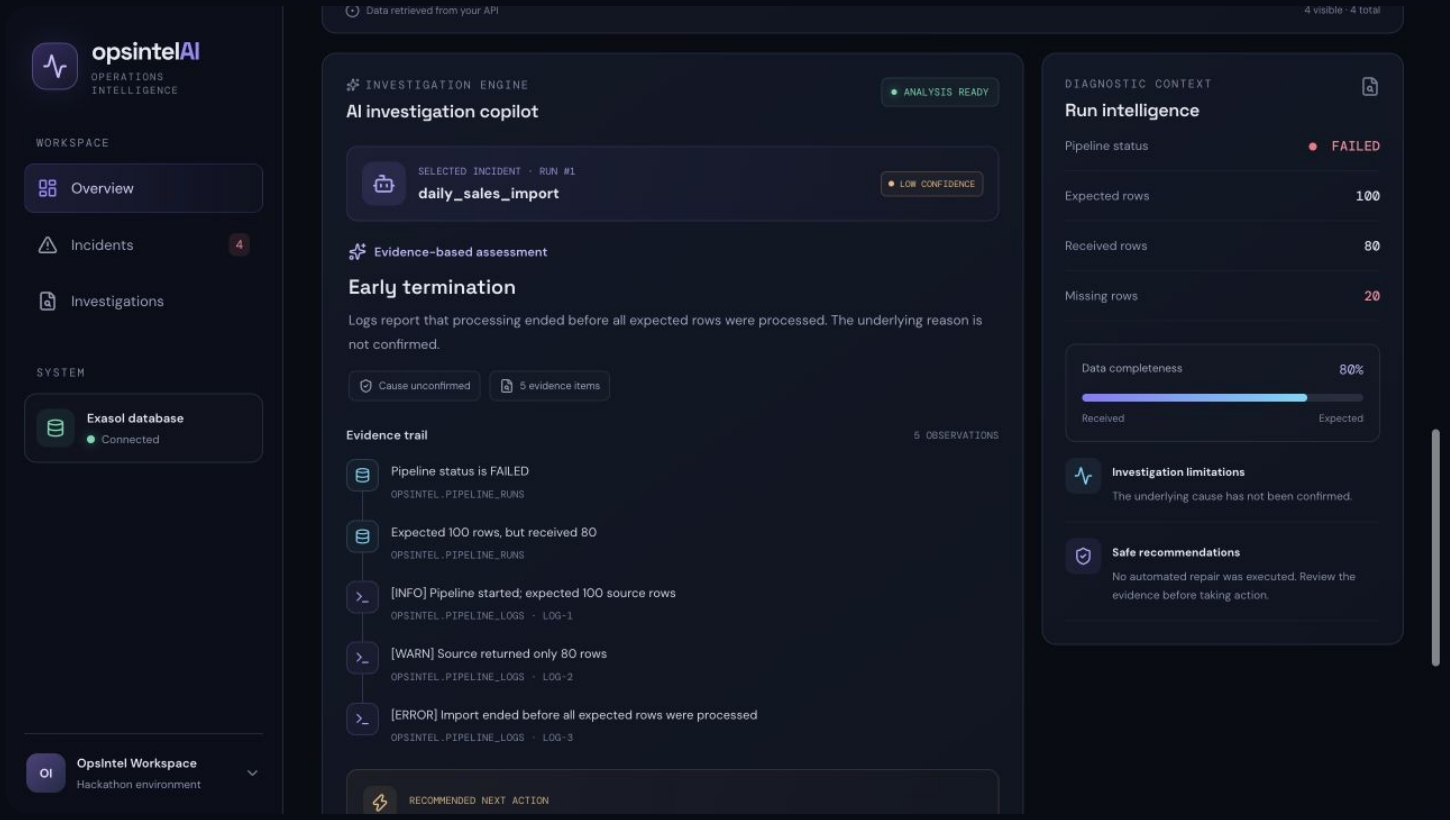
Selected run

daily_sales_import

OBSERVATION

Expected: 100 rows
Received: 80 rows
Missing: 20 rows

EVIDENCE TRAIL
PIPELINE_RUNS: status and row counts
LOG-1: expected 100 source rows
LOG-2: source returned only 80 rows
LOG-3: import ended before completion



CANDIDATES

Source shortfall or early termination
Confidence: low

CONFIRMATION

None
Human verification required

WHY EXASOL

Exasol computes incident context before FastAPI responds

ANALYTICAL FEATURES

01 Previous run status with LAG

02 Rolling actual-row baseline

03 Rolling failure rate

04 Missing rows and completeness

05 FastAPI reads the view directly

QUERY USED BY /analytics/trend

```
SELECT TO_CHAR(RUN_TIME, 'YYYY-MM-DD') AS RUN_DAY,
       COUNT(*) AS TOTAL_RUNS,
       SUM(CASE WHEN RUN_STATUS = 'FAILED'
         OR EXPECTED_ROWS > ACTUAL_ROWS THEN 1 ELSE 0 END)
         AS INCIDENT_COUNT,
       ROUND(AVG(COMPLETENESS_PCT), 1) AS AVG_COMPLETE_PCT
FROM OPSINTEL.PIPELINE_HEALTH_FEATURES
GROUP BY TO_CHAR(RUN_TIME, 'YYYY-MM-DD')
```

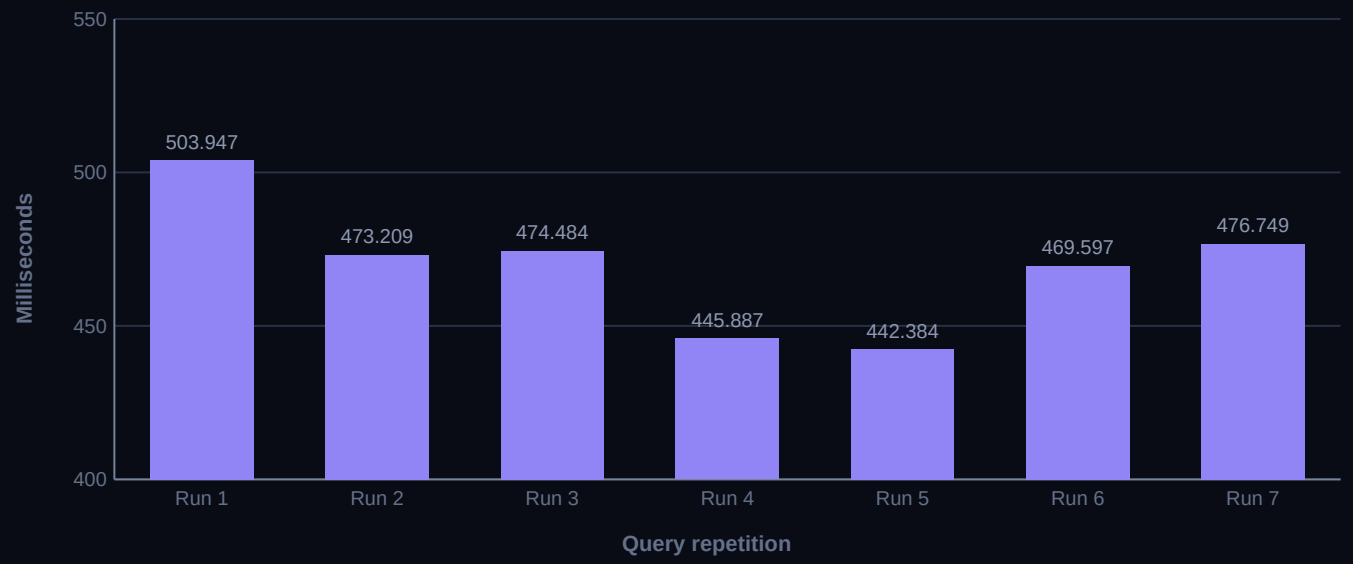
ACTUAL OUTPUT FROM LOCAL EXASOL

2026-10-15 5 runs 4 incidents
56.0% average completeness 16.0% rolling failure rate

FastAPI returns these view rows directly. Exasol keeps the window calculations beside the run history, so Python does not rebuild them.

MEASURED EVIDENCE

1,000,000 synthetic run rows: 473 ms median and 504 ms p95



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predefined synthetic challenge cases

3.025 s

1,000,000-row import

473 ms

median analytical query

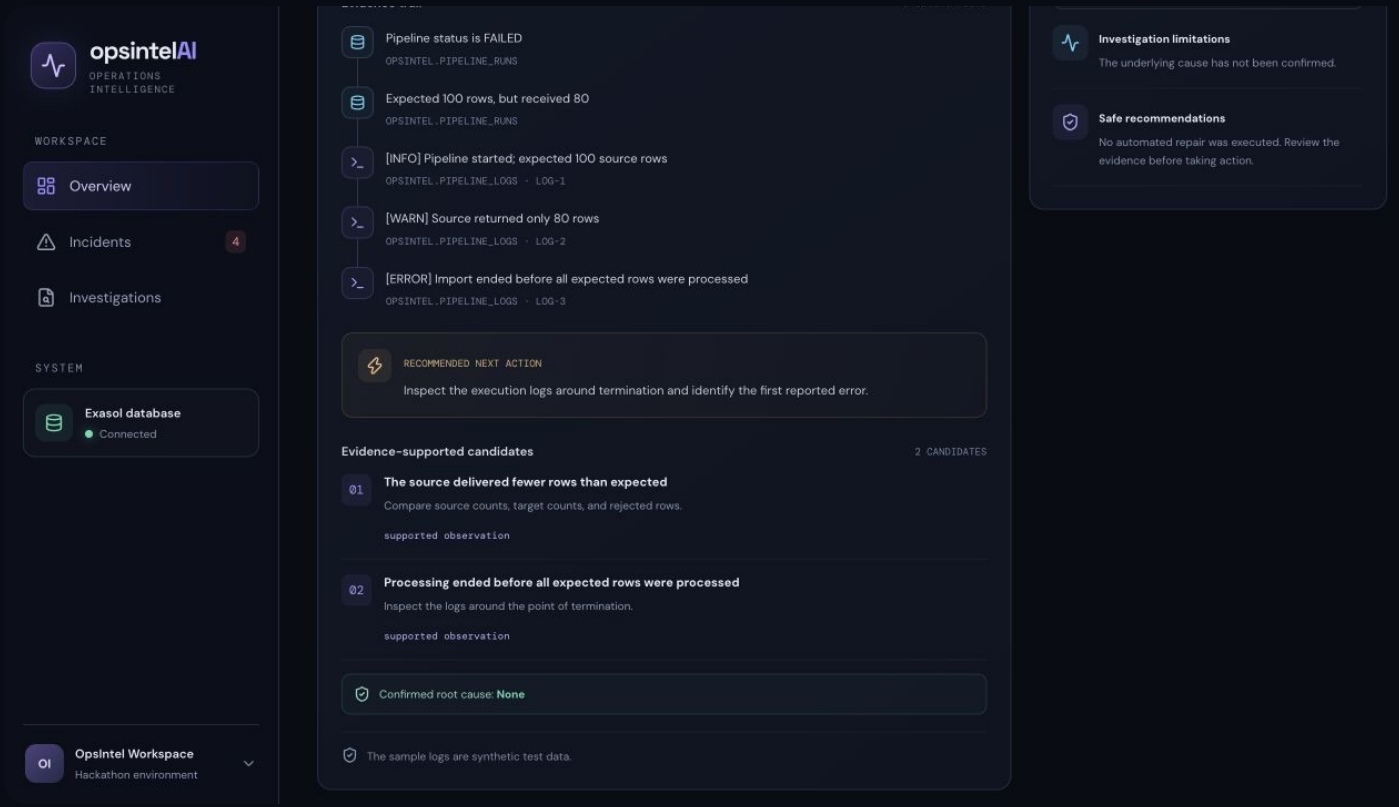
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automated tests passed

Records: reports/exasol-benchmark.json. Reproduce with scripts/benchmark_exasol.py, pytest -q, and tests/run_evaluation.py. Local single-user, 7 repeats. No concurrency claim.

TRUST AND SAFETY

The product is designed to abstain before it overclaims



OBSERVATION

Source-backed facts

CANDIDATE

Evidence-supported possibility

CONFIRMATION

Reserved for human verification

INSUFFICIENT EVIDENCE EXAMPLE
Worker exited with code 71. No diagnostic details. Result: unclassified. Optional summary cannot confirm a cause or run a repair.

VALIDATION ROADMAP

What is proven, and what must be proven next

PROVEN NOW

01 Live Exasol-backed demo

02 One-million-row benchmark

03 49 automated tests

04 Optional structured LLM summary

DEFINED, NOT COMPLETED

01 Reviewer holdout protocol

02 Frozen-commit process

03 Human impact study protocol

04 LLM summary quality evaluation

PRODUCTION WORK

01 Orchestrator connectors

02 Authentication and authorization

03 Concurrent workload testing

04 Audited retention and monitoring

No independent holdout result or measured productivity improvement is claimed.

OPSINTEL AI

Follow the evidence. Investigate faster. Confirm carefully.

github.com/Anafa-thabassum/opsintel-ai

Live demo: one incident, five observations, one safe next action